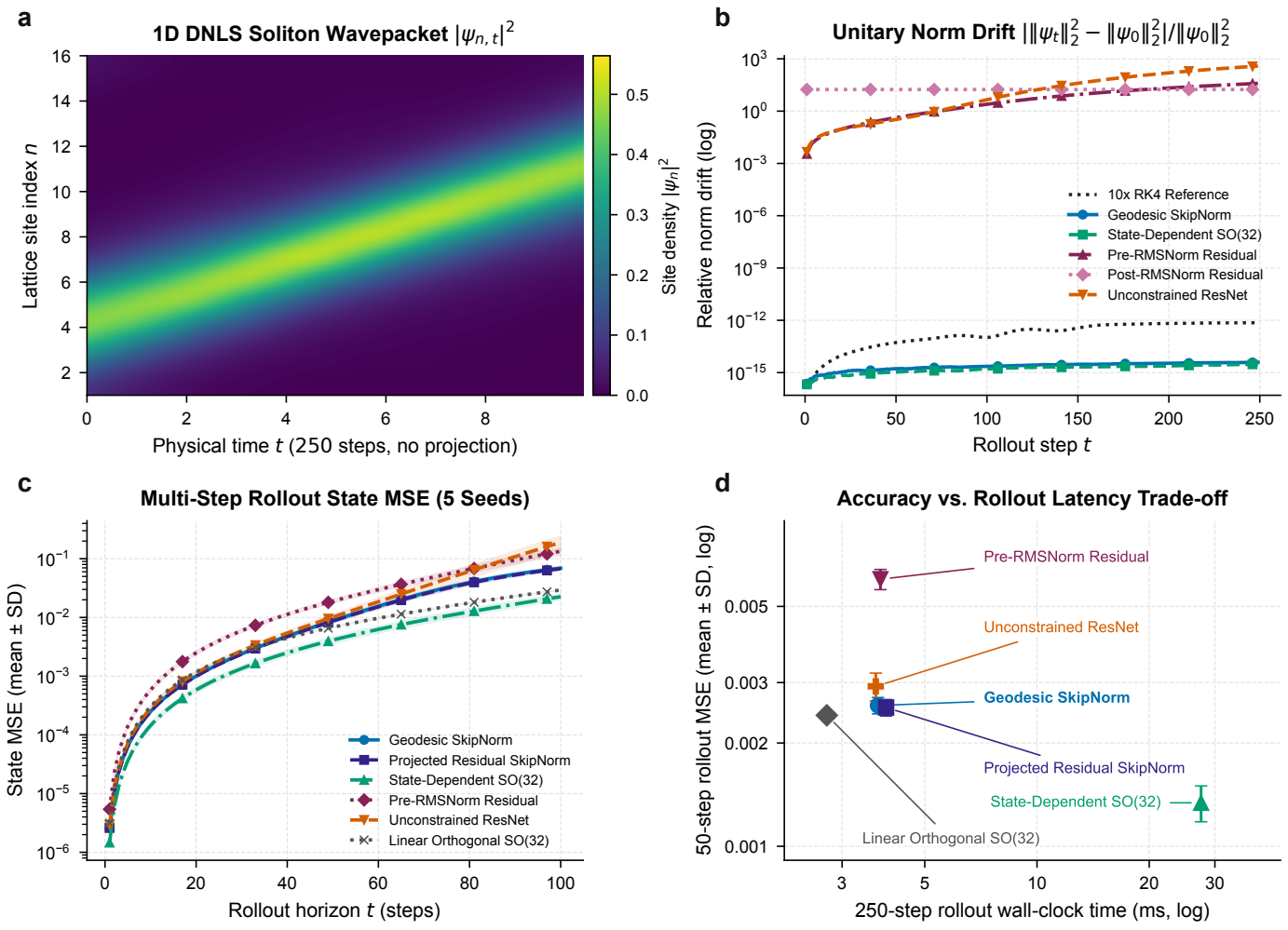

a
1D DNLS Soliton Wavepacket $|\psi_{n,t}|^2$
Lattice site index n
Physical time t (250 steps, no projection)
Site density $|\psi_n|^2$
b
Unitary Norm Drift $|\|\psi_t\|_2^2 - \|\psi_0\|_2^2|/\|\psi_0\|_2^2$
Relative norm drift (log)
Rollout step t
10x RK4 Reference
Geodesic SkipNorm
State-Dependent SO(32)
Pre-RMSNorm Residual
Post-RMSNorm Residual
Unconstrained ResNet
c
Multi-Step Rollout State MSE (5 Seeds)
State MSE (mean ± SD)
Rollout horizon t (steps)
Geodesic SkipNorm
Projected Residual SkipNorm
State-Dependent SO(32)
Pre-RMSNorm Residual
Unconstrained ResNet
Linear Orthogonal SO(32)
d
Accuracy vs. Rollout Latency Trade-off
50-step rollout MSE (mean ± SD, log)
250-step rollout wall-clock time (ms, log)
Pre-RMSNorm Residual
Unconstrained ResNet
Geodesic SkipNorm
Projected Residual SkipNorm
State-Dependent SO(32)
Linear Orthogonal SO(32)